Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 7 [2024 P2]

Test Date: 27th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. Let P be the foot of perpendicular drawn from a point Q. where Q lies on the line $x+y+4=0$ and P lies on the line $5x+y=0$. If P is the image of point $R(12,0)$ about the line $x+y=2$, then coordinates of Q is

A) $\left(\frac{14}{3},\frac{-26}{3}\right)$ **B)** $\left(\frac{11}{3},\frac{-23}{3}\right)$ **C)** $\left(\frac{16}{3},-\frac{28}{3}\right)$ **D)** $(6,-10)$

2. Area of region bounded by curve $y=[x]+\sqrt{x-[x]}, y=0, x=0, x=6$ is equal to (in sq. units)

A) 6 **B)** 13 **C)** 19 **D)** 23

3. A line passing through origin cuts the lines $\frac{x+1}{1}=\frac{y-2}{1}=\frac{z-1}{-2}$ and $\frac{x-1}{1}=\frac{3y-8}{6}=\frac{z+3}{-1}$ at A and B respectively, then the square of length of AB is equal to

A) 3 **B)** 6 **C)** 9 **D)** 12

4. Let S be the set of all possible 2×2 matrices A with integer entries such that $AA^T=2I$ where $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, then number of such matrices in S such that $\det(A-I)=-1$ (where $\det(A)$ denotes the determinant of matrix A) is

A) 2 **B)** 4 **C)** 6 **D)** 8

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let $A=\begin{pmatrix}2 & 1 & 3\\ 1 & 1 & 2\\ 3 & 1 & 1\end{pmatrix}$ and $A^{-1}=\alpha A^2+\beta A+\gamma I$, then which of the following holds good?

A) $\alpha+\beta=1$ **B)** $3(\alpha+\beta+\gamma)=10$

C) $8(\beta+\gamma)=11$ **D)** $8(\alpha+\gamma)=3$

6. $f(x)=\begin{cases}3x^5+2x^3, & x\in \text{ Rational}\\ 0, & x\in \text{ Irrational}\end{cases}$, then

A) $f(x)$ is continuous exactly at one point

B) $f(x)$ is discontinuous $\forall x\in R$

C) $f(x)$ is non-differentiable $\forall x\in R$

D) $f(x)$ is differentiable exactly at one point

7. If $w=\dfrac{z-1}{z-2}, |z|=1, z, w\in C$, then:

A) $|w-3|_{\min}=\dfrac{7}{3}$ **B)** $|w-3|_{\max}=3$

C) $(\text{Arg}(w))_{\max}=\dfrac{\pi}{4}(-\pi<\text{Arg}(w)<\pi)$ **D)** $(\text{Arg}(w))_{\min}=0(-\pi<\text{Arg}(w)<\pi)$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

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8. Let $a,b,c>0$ and $\dfrac{1}{1+a^4}+\dfrac{1}{1+b^4}+\dfrac{1}{1+c^4}=1$, then the minimum value of $a^2b^2c^2$ is k then value of k^2 is

9. The value of $12\cdot\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sqrt{1+\dfrac{k}{n^2}+\dfrac{k^2}{n^3}}-1\right)$ is equal to

10. Let z_1,z_2,z_3 and z_4 be the roots of the equation $z^4-8iz^3-24z^2+32iz+15-i=0$ and $P(z)$ be a point on complex plane, then the minimum value of $\sum\limits_{i=1}^{4}|z-z_i|$ is 2^x, then the value of $[x]$ is equal to (where $[\cdot]$ represent the greatest integer function)

11. The value of $270\sum\limits_{a=1}^{\infty}\sum\limits_{b=1}^{\infty}\sum\limits_{c=1}^{\infty}\dfrac{ab(3a+c)}{4^{a+b+c}(a+b)(b+c)(c+a)}$ is

12. Let n be a positive integer. If the number of integers in the domain of the function $f(x)=\ln\left((1-x)(x-n)\right)$ is $2n-11$, then $\cot^{-1}(\cot n)$ is equal to k then value of $[k]$ is: [Use $\pi=3.14$] (where $[\cdot]$ represent the greatest integer function)

13. From a fixed point A, three normal are drawn to the parabola $y^2=4ax$ at the points P,Q and R. Two circles C_1 and C_2 be drawn on AP and AQ as diameters. If the slope of the common chord of circles C_1 and C_2 be m_1 and the slope of the tangent to the parabola at the point R be m_2, then $|16m_1m_2|$ is equal to

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Let set A contain all the 3×3 matrices whose entries are $0,1$ & -1. Two of these entries are 1, two are -1 & five are 0.

On the basis of above information, answer the following questions:

14. Number of skew-symmetric matrix in set A is

15. Number of symmetric matrix in set A which is non-singular is

PARAGRAPH-II

Let $f:R\to R$ and $g:R\to R$ such that

$f(x)=max\left\{|x|,|x-1|,|x-2|,.....,|x-2018|\right\}$

$g(x)=min\left\{|x|,|x-1|,|x-2|,.....,|x-2018|\right\}$ then:

16. The value of $\int_0^{2018} f(x)dx$ is k then value of $\sqrt{3k}$ is equal to

17. The value of $\dfrac{\int_0^{2018} f(x)dx}{\int_0^{2018} g(x)dx}$ is

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

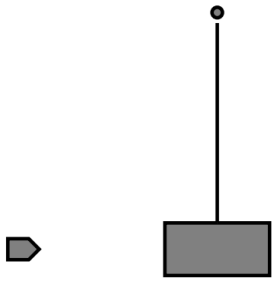
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18. A wooden block (mass M) is hung from a peg by a massless rope. A speeding bullet (with mass m and initial speed v_0) collides with the block at time $t = 0$ and embeds in it. Let *S* be the system consisting of the block and bullet. Which quantities are conserved between $t = -10\ s$ and $t = +10\ s$?



A) The horizontal component of the linear momentum of *S*.

B) The mechanical energy of *S*.

C) The angular momentum of *S* as measured about a perpendicular axis through the peg.

D) None of the above are conserved.

19. Four objects are placed at rest at the top of an inclined plane and allowed to roll without slipping to the bottom in the absence of rolling resistance and air resistance.

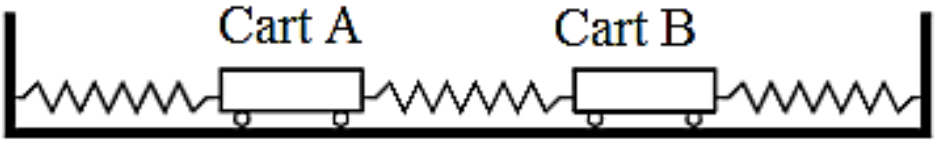
• Object *A* is a solid brass ball of diameter d.
• Object *B* is a solid brass ball of diameter 2d.
• Object *C* is a hollow brass sphere of diameter d.
• Object *D* is a solid aluminium ball of diameter d. (Aluminium is less dense than brass.)

The balls are placed so that their centres of mass all travel the same distance. In each case, the time of motion T is measured. Which of the following statements is correct?

A) $T_B > T_C > T_A = T_D$ **B)** $T_A = T_B = T_C > T_D$

C) $T_B > T_A = T_C = T_D$ **D)** $T_C > T_A = T_B = T_D$

20. Two identical carts A and B each with mass *m* are connected via a spring with spring constant *k*. Two additional springs, identical to the first, connect the carts to two fixed points. The carts are free to oscillate under the effect of the springs in one dimensional frictionless motion.

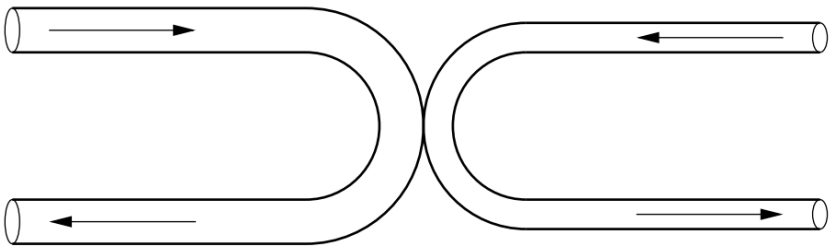


Under suitable initial conditions, the two carts will oscillate in phase according to $x_A(t) = x_0 \sin\omega_1 t = x_B(t)$ where x_A and x_B are the locations of carts A and B relative to their respective equilibrium positions. Under other suitable initial conditions, the two carts will oscillate exactly out of phase according to $x_A(t) = x_0 \sin\omega_2 t = -x_B(t)$

Determine the ratio ω_2 / ω_1

A) $\sqrt{3}$ **B)** 2 **C)** $2\sqrt{2}$ **D)** 3

21. Two streams of water flow through the U-shaped tubes shown. The tube on the left has cross-sectional area A, and the speed of the water flowing through it is *v*; the tube on the right has cross-sectional area A' = A/2. If the net force on the tube assembly is zero, what must be the speed v_0 of the water flowing through the tube on the right? Neglect gravity, and assume that the speed of the water in each tube is the same upon entry and exit.



A) $1/2v$ **B)** v **C)** $\sqrt{2}v$ **D)** $2v$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

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22. A particle of mass 2.00 kg moves under a force given by $\vec{\mathbf{F}} = -(8.00\ \text{N/m})(x\hat{\mathbf{i}} + y\hat{\mathbf{j}})$

where $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are unit vectors in the x and y directions. The particle is placed at the origin with an initial velocity $\vec{\mathbf{v}} = (3.00\ \text{m/s})\hat{\mathbf{i}} + (4.00\ \text{m/s})\hat{\mathbf{j}}$.

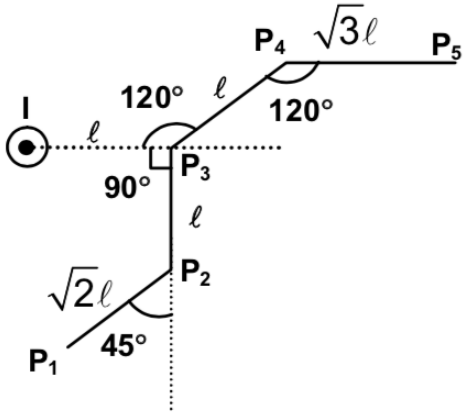
A) particle first return to the origin at $t = 1.57$ s

B) the maximum distance between the particle and the origin is 5.00 m

C) magnitude of maximum acceleration of particle is $10\ \text{m/s}^2$

D) the maximum distance between the particle and the origin is 2.50 m

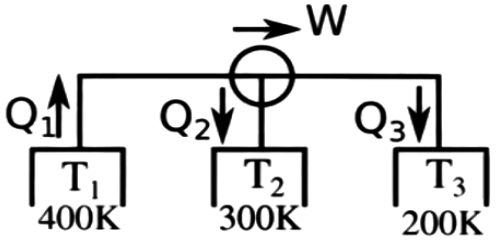
23. A straight conducting wire of infinite length lies perpendicular to the plane of the paper and current of magnitude 'I' is flowing out to the paper as shown in figure. Then choose the correct option(s).



A) The value of $\int_{P_1P_2} \vec{B}\cdot\vec{dl} = \frac{\mu_0 I}{8}$. **B)** The value of $\int_{P_1P_2} \vec{B}\cdot\vec{dl} = \int_{P_2P_3} \vec{B}\cdot\vec{dl}$.

C) The value of $\int_{P_3P_4} \vec{B}\cdot\vec{dl} = 2\int_{P_4P_5} \vec{B}\cdot\vec{dl}$. **D)** The value of $\int_{P_1P_2P_3P_4P_5} \vec{B}\cdot\vec{dl} = \frac{7\mu_0 I}{24}$.

24. During a number of complete cycles, a reversible engine (shown by a circle) absorbs 1200 joule from reservoir at 400 K and performs 200 joule of mechanical work then:



A) The reservoir at temperature T_2, absorbs 200J of heat while the reservoir at temperature T_3, loses 1200J of heat.

B) The reservoir at temperature T_2, absorbs 1200J of heat while the reservoir at temperature T_3, loses 200J of heat.

C) Change in entropy of reservoir at temperature T_2, will be 4J/K

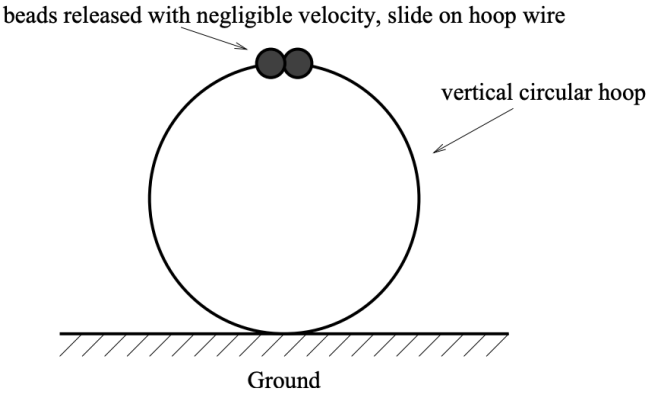
D) Change in entropy of reservoir at temperature T_3, will be -1J/K

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..

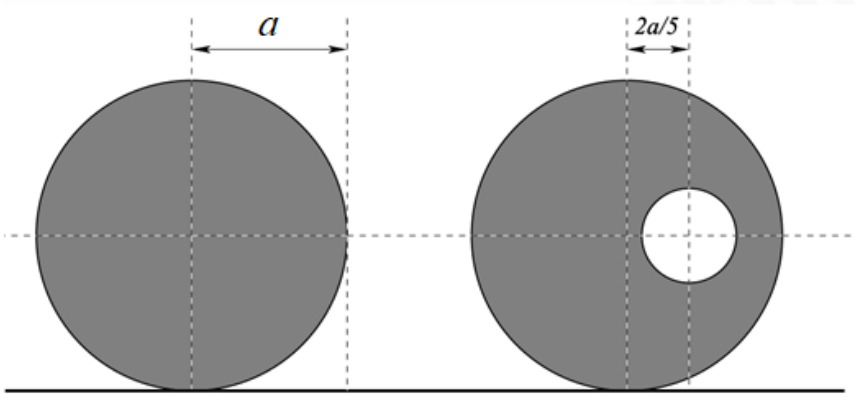
• For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
• Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases..*

25. Two beads, each of mass m, are free to slide on a rigid, vertical hoop of mass m_h. The beads are threaded on the hoop so that they cannot fall off of the hoop. They are released with negligible velocity at the top of the hoop and slide down to the bottom in opposite directions. The hoop remains vertical at all times. The maximum value of the ratio m/m_h such that the hoop always remains in contact with the ground is x. fill the value of $6x$. (Neglect friction).



26. Two identical charged particles having charge $10\mu C$ and mass 100 gm each are released on a rough horizontal plane surface at a distance of 1 m from each other. Coefficient of friction between the particles and horizontal plane surface is 0.1 and acceleration due to gravity is 10 m/s^2. Find Maximum speed (in SI unit) acquired by each particle.

27. A uniform cylinder of radius a originally has a weight of 80 N. After an off-axis cylinder hole at 2a/5 was drilled through it, it weighs 65 N. The axes of the two cylinders are parallel and their centres are at the same height.

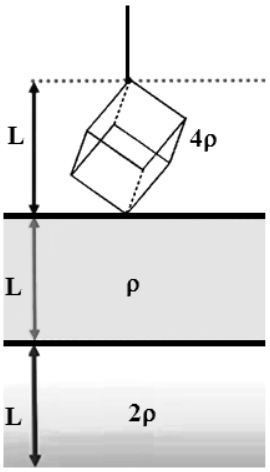


A force T is applied to the top of the cylinder horizontally. In order to keep the cylinder at rest, the magnitude of the force T is (in N):

28. A charged particle is moving with a velocity $\vec{v} = -\hat{i} + \hat{j} - \hat{k}$ and the electric field produced by it at a distant point is $\vec{E} = +\hat{i} + \hat{j} + \hat{k}$. The magnetic field produced at the

same point due to the same charged particle is $K\mu_0\varepsilon_0$. Find the value of $(K)^2$. (All quantities are in S.I. units, take $\sqrt{2}=1.4$ and $\sqrt{3}=1.7$)

29. A cube having body diagonal of length L and density 4ρ (Mass 4M) is being slowly lowered into a very wide pool containing two liquids with the help of a thread as shown. Magnitude of the work done by the tension in process from the moment the bottom vertex of the cube is touching surface of the top liquid to the moment when the bottom vertex just touches the bottom of lower liquid is *x*MgL. Fill the value of *x*.



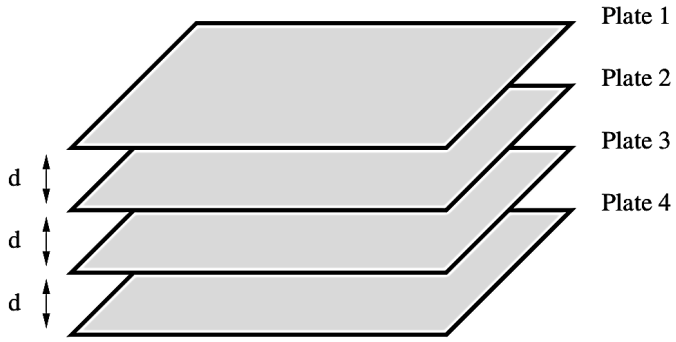
30. A light bulb has a solid cylindrical filament of length L and radius a, and consumes power P. You are to design a new light bulb, using a cylindrical filament of the same material, operating at the same voltage, and emitting the same spectrum of light, which will consume power *nP*. The length (L_1) and radius (a_1) of the new filament are $L_1 = n^x L$ and $a_1 = n^y a$. Find *y*/*x*. Assume that the temperature of the filament is approximately uniform across its cross-section; the filament doesn't emit light from the ends; and energy loss due to convection is negligible.

SECTION-IV
(PARAGRAPH QUESTIONS)

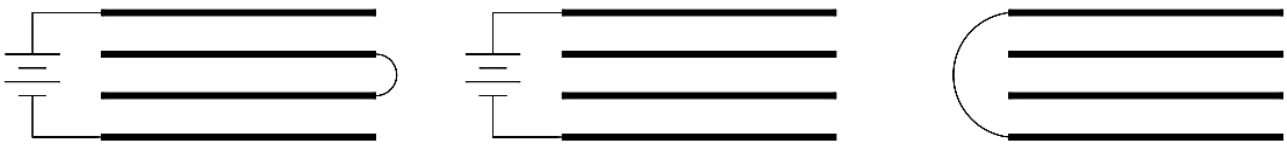
- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Four square metal plates of area A are arranged at an even spacing *d* as shown in the diagram. (Assume that $A >> d^2$).



Plates 1 and 4 are first connected to a voltage source of magnitude V_0, with plate 1 positive; plates 2 and 3 are then connected together with a wire. The wire is subsequently removed. Finally, the voltage source attached between plates 1 and 4 is replaced with a wire. The steps are summarized in the diagrams below.



Step 1 Step 2 Step 3

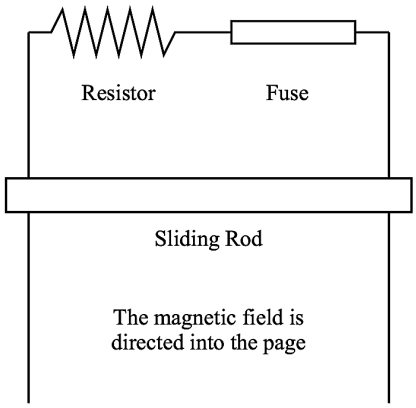
31. The resulting potential difference $(\Delta V_{12} = V_1 - V_2)$ between plates 1 and 2 is xV_0. Fill the value of x.

32. The resulting potential difference ΔV_{23} between plates 2 and 3 is yV_0. Fill the value of y.

PARAGRAPH-II

One pair of ends of two long, parallel wires are connected by a resistor, R = 0.25 Ω, and a fuse that will break instantaneously if 5 amperes of current pass through it. The other pair of ends are unconnected. A conducting rod of mass m is free to slide along the wires under the influence of gravity. The wires are separated by 30 cm, and the rod starts out 10 cm from the resistor and fuse. The whole system is placed in a uniform, constant magnetic field of B = 1.2 T as shown in the figure. The resistance of the rod and the wires is negligible. When the rod is released it falls under the influence of gravity, but never loses contact with the long parallel wires.

(Take $g = 10\,\text{m/s}^2$)



33. What is the smallest mass (in kg) needed to break the fuse?

34. How fast (in m/s) is the mass moving when the fuse breaks?

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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35. Which orbital overlaps among the following is not possible in 'pentadiene' molecule?

A) $sp^3 - sp^3$ **B)** $sp^2 - sp^2$ **C)** $sp - sp$ **D)** $sp^2 - sp^3$

36. For the four covalent bonds; H—F, H—O, F—O and O═O, the following bond energies are listed (in random order)

44 kcal/mole
111 kcal/mole
119 kcal/mole
135 kcal/mole

If the magnitude of ΔH for the reaction $F_2O(g) + H_2O(g) \rightarrow O_2(g) + 2HF(g)$ is 79 kcal, identify the incorrect statement(s)

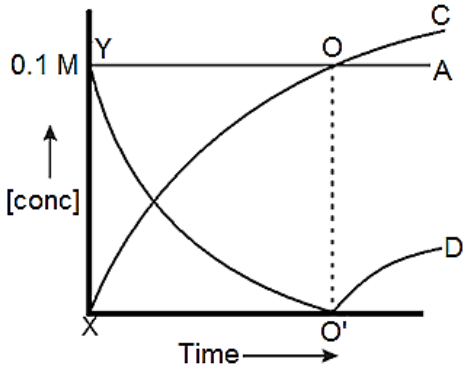
A) F—O bond will be the weakest of the lot

B) O═O bond will be the strongest of the lot

C) H—F bond will be stronger than H—O bond

D) the reaction mentioned above is exothermic

37. In a solution of 0.1M $AgNO_3$, sodium chloride salt is slowly added. The concentrations of different ions (before the saturation point of strong electrolytes) can be represented as (graph not according to scale and assume there is no change in volume during process)



Select the correct statement

A) XC represents the $[Ag^+]$

B) O'D represents the $[Cl^-]$ after the complete precipitation of AgCl

C) YA represents the $[Na^+]$

D) YO' represents the $[NO_3^-]$

38. If clockwise spin were represented by the number 9 and anti clockwise spin by the number 6, the 2p subshell of oxygen atom in ground state can attain a maximum numerical value of (assume all rules of electron filling apply)

A) 6966 **B)** 9699 **C)** 9996 **D)** 9999

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Which of the following atoms are less electronegative than phosphorus:

A) *Si* **B)** *Sn* **C)** *S* **D)** *Se*

40. A hydrocarbon (W) has molecular formula $C_{13}H_{24}$. Upon hydrogenation of it's pi-bond it converts to 4-Cyclopentyl-4-methylheptane. Which of the following can be a characteristic for the possible structure of (W):

A) presence of chiral carbon outside the cyclic ring

B) absence of geometrical isomerism in the molecule

C) presence of chiral carbon upon the cyclic ring

D) simultaneous presence of chiral centre and geometrical isomerism in the molecule

41. Bachmann and Ferguson calculated the migratory aptitude of various aryl groups (listed on the left) and came up with certain numbers (listed on the right)

Aryl group	Migratory aptitude
o-anisyl	0.3
p-anisyl	0.7
p-toluyl	1.95
m-toluyl	15.7
p-chlorophenyl	500

Due to typist's mistake the numbers in the above chart have got rearranged such that only one number written infront of its group is actually right. All these number are calculated w.r.t. phenyl ring whose migratory aptitude is taken as 1.

Identify the correct matching(s)

A) p – toluyl : 15.7 **B)** m – toluyl : 0.7

C) p – anisyl : 500 **D)** o – anisyl : 0.3

SECTION-III
(NON-NEGATIVE INTEGER)

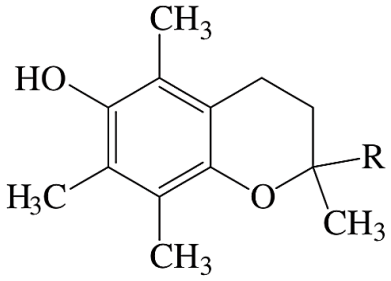
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

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42. Solubility of $Pb(OH)_2(s)$ in pure water is $10^{-4}M$. How many times the solubility of $Pb(OH)_2$ will increase in buffer of $pH = 10$? [Consider Temperature = $25^{\circ}C$]

43. Shown below is the structure of Vitamin-E having molar mass of 430 grams.



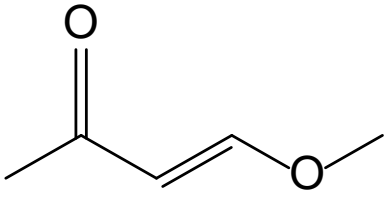
(R = alkyl chain)

The number of carbon atoms in the alkyl chain is _____

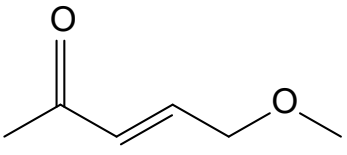
44. A crystalline solid is made of ccp packing of 'X' atoms with O atoms occupying all the edge centres and Y atoms occupying all the tetrahedral voids created by X atoms. The formula of its unit cell is $X_aY_bO_c$. If all the atoms along any one C_3 axis of symmetry are removal from the unit cell; the simplest formula of the resulting solid is $X_dY_bO_a$.

Calculate the value of $a+b+c+d$.

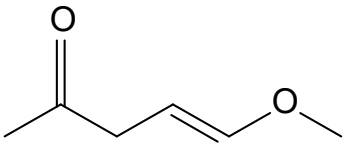
45. Select all the compounds having higher rotational energy barrier across $C=C$ as compared to the following molecule and report the sum of their serial number as your response.

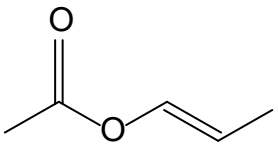


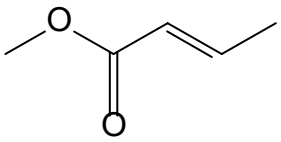
1

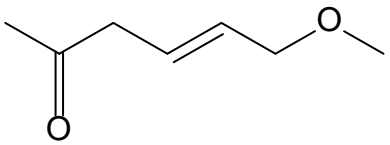


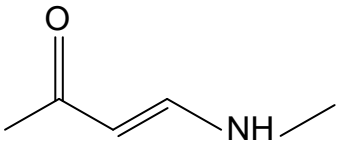
2



3 

4 

5 

6 

46. 2 moles $C(g)$ and 4 moles $D(g)$ are mixed together in a sealed 1 litre vessel and allowed to attain equilibrium as $A(g)+2B(g) \rightleftharpoons C(g)+D(g)$. K_c for the above equilibria is 10^{-8} lit/mole. If the equilibrium concentration for $C(g)$ is found to be $x \cdot y \times 10^{-z}$ (x point y into 10 raised to minus z), calculate the value of $x+y-z$.

47. For the decomposition reaction $X \longrightarrow Y$, 50% reaction occurs in 10 min. and the next 75% reaction takes 1 hour to finish. Tabulate the order of the reaction.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Carefully study the given segment of the modern periodic table:

		1	2
		3	Si
Cu	4	Ga	5
6	7	8	9
10	Hg	11	12
13	14	15	Fl

Symbols of a few elements are mentioned in the appropriate box while the rest are hidden inside boxes numbered 1 to 15.

48. If the atomic number of each of the hidden elements were divided by their box number, how many natural numbers would show up?

49. If the stable oxidation states of elements hidden inside boxes 4, 8 and 12 are X, Y and Z respectively in their compound, find the value of $X \times Y \times Z$

PARAGRAPH-II

Proline was first isolated in 1900 by Richard Willstatter. In the year that followed, Emil Fischer isolated proline from casein, a milk protein. The amino acid gets its name from the heterocyclic pyrrolidine unit it contains. It is a conditionally essential amino acid that helps build collagen and supports healthy blood pressure and arterial elasticity.

50. If for proline, $pKa_1 = 2$ and $pKa_2 = 10.6$; then at $pH = 7$ proline will exists as:

1. overall neutral structure with no charge sites

2. overall positively charged entity

3. overall negativity charged entity

4. overall neutral structure with equal number of oppositely charged sites

Select the correct number code from above and multiply it by the molecular weight of N-methylproline (molar mass 129) to arrive at the answer.

51. When Proline undergoes Ninhydrin test, what colour is seen?



Select the correct colour code from above and multiply it by the number of carbons atoms present in the pyrrolidine heterocycle to arrive at the answer.